

Week 1 – Hoofdstuk 1 & 2: Vectoren, lijnen, vlakken en het inproduct

Studiegids uit de HC-aantekeningen (dictaat-nummers ter referentie)

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Lecture framing: “Voorbereidingen voor lineaire algebra. H1 + H2, meeste herhaling VWO.” So this week is mostly a refresher, but the notation and the way of thinking in sets ($\{ \dots \mid \lambda \in \mathbb{R} \}$) are used in every later week.

1. The intuitive vector (Def. 1.1.1)

A **vector** is a *translation in space* (een verschuiving). It has two pieces of data:

- a **richting** (direction),
- a **grootte** / **lengte** (magnitude), written $|\mathbf{v}|$.

If the arrow goes from point A to point B we write $\mathbf{v} = \overrightarrow{AB}$. Two arrows that are parallel, equally long and point the same way are the *same* vector.

2. Bewerkingen (operations) — Def. 1.1.2–1.1.4, rules in Stelling 1.1.5

#	Operation	Picture / meaning
1	Optelling $\mathbf{a} + \mathbf{b}$	Put the tail of $\mathbf{b}$ at the head of $\mathbf{a}$; the sum is the arrow from the start of $\mathbf{a}$ to the end of $\mathbf{b}$ (parallelogram rule).

#	Operation	Picture / meaning
2	Scalaire vermenigvuldiging $\lambda \mathbf{a}, \lambda \in \mathbb{R}$	Same direction, length scaled by λ (e.g. $2\mathbf{a}$, $2.7\mathbf{a}$); $\lambda < 0$ flips the direction ($-\mathbf{a}$).
3	Verschilvector $\mathbf{b} - \mathbf{a}$	The vector such that $\mathbf{a} + (\mathbf{b} - \mathbf{a}) = \mathbf{b}$: the arrow from the head of $\mathbf{a}$ to the head of $\mathbf{b}$.
4	Nulvector $\mathbf{0}$	Has length 0.

“Scalaire” just means *a number from* $\mathbb{R}$. The rekenregels (commutativity, associativity, distributivity) are listed in Stelling 1.1.5 of the dictaat (p. 7); you may use them without proof.

3. Doing geometry: oorsprong + assenstelsel (§1.2, §1.3)

To actually *compute* with vectors we make two choices:

1. **Oorsprong** O : every vector is drawn starting at O . Then a point P is identified with the vector $\overrightarrow{OP}$, and O itself with $\mathbf{0}$.
2. **Assenstelsel**: choose an *ordered* set of vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ (for 3-dimensional space) that are
 - of length 1, and
 - **onderling loodrecht** (mutually perpendicular).

With respect to (t.o.v.) $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ every vector can be written as

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + x_3\mathbf{e}_3 = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad x_i \in \mathbb{R}.$$

The numbers x_1, x_2, x_3 are the **coördinaten** of $\mathbf{x}$ t.o.v. $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ (Def. 1.3.1). The column of coordinates is an element of $\mathbb{R}^3$ (Def. 1.3.2). In a line of text the dictaat writes $(x_1, x_2, x_3)^t$ for the column.

4. Points, lines and planes as *sets* of vectors (§1.2, §1.4)

The key habit of this course: a geometric object is a **verzameling** (set) of vectors, written as { formula | parameter ranges over $\mathbb{R}$ }. Read | as “waarvoor geldt” (for which holds).

Lines

Case	Parametrisation	Names
VB1: line l through $\mathbf{0}$	$l = \{\lambda \mathbf{a} \mid \lambda \in \mathbb{R}\}$	$\mathbf{a}$ is the richtingsvector
VB2: line m <i>not</i> through $\mathbf{0}$	$m = \{\mathbf{p} + \lambda \mathbf{a} \mid \lambda \in \mathbb{R}\}$	$\mathbf{p}$ is the steunvector , $\mathbf{a} \neq \mathbf{0}$ the richtingsvector

This is Def. 1.2.1. Line through two points P and Q (with vectors $\mathbf{p}, \mathbf{q}$):

$$\{\mathbf{p} + \lambda(\mathbf{q} - \mathbf{p}) \mid \lambda \in \mathbb{R}\}$$

(steunvector $\mathbf{p}$, direction = the difference vector $\mathbf{q} - \mathbf{p}$).

Planes

Case	Parametrisation
VB3: plane V through $\mathbf{0}$	$V = \{\lambda \mathbf{a} + \mu \mathbf{b} \mid \lambda, \mu \in \mathbb{R}\}$
VB4: plane W not through $\mathbf{0}$	$W = \{\mathbf{p} + \lambda \mathbf{a} + \mu \mathbf{b} \mid \lambda, \mu \in \mathbb{R}\}$

Requirement: the two richtingsvectoren $\mathbf{a}, \mathbf{b}$ must **not lie on one line** (they are *onafhankelijk*, Def. 1.2.4: neither is a scalar multiple of the other). This is Def. 1.2.5. Sketch from the lecture: with $\lambda \mathbf{a}$ and $\mu \mathbf{b}$ you can reach *every* point of the plane, which is what “parametriseren” means.

Plane through three points P, Q, R (not on one line):

$$V = \{\mathbf{p} + \lambda(\mathbf{q} - \mathbf{p}) + \mu(\mathbf{r} - \mathbf{p}) \mid \lambda, \mu \in \mathbb{R}\}.$$

*Steunvector and richtingsvectoren are **not unique**: any point of the object can serve as steunvector, any non-zero multiple of a direction works.*

5. Length of a vector in $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^n$

Space	Vector	Length
1-dim, $\mathbb{R}$	$\mathbf{x} = x_1 \mathbf{e}_1 = (x_1)$	$ \mathbf{x} = x_1 = \sqrt{x_1^2}$ (absolute value)
2-dim, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$	$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$	$ \mathbf{x} = \sqrt{x_1^2 + x_2^2}$ (Pythagoras)
3-dim, $\mathbb{R}^3$	$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3$	$ \mathbf{x} = \sqrt{x_1^2 + x_2^2 + x_3^2}$ (Pythagoras applied twice : first in the floor, then up)
n -dim, $\mathbb{R}^n$ ($n \in \mathbb{N}, n > 0$)	$(x_1, \dots, x_n)^t$ t.o.v. $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$	$\sqrt{x_1^2 + \dots + x_n^2}$

Why bother with $n > 3$? Two lecture examples:

- A particle: $(x_1, x_2, x_3, v_1, v_2, v_3, t, T)^t$ = position, velocity, time, temperature $\rightarrow$ a vector in $\mathbb{R}^8$.
- Ten factories, x_i = production of factory i in a year $\rightarrow (x_1, \dots, x_{10})^t \in \mathbb{R}^{10}$.

Def. 2.5.1: $\mathbb{R}^n$ is the set of ordered n -tuples of reals; addition and scalar multiplication go coordinate-wise.

6. The inner product (inproduct / dotproduct) — plan in three steps

The lecturer’s plan:

1. Define the inner product on $\mathbb{R}^2$ and $\mathbb{R}^3$ **geometrically**.
2. Express it **in coordinates**.
3. Use the coordinate formula to **define** it on $\mathbb{R}^n$ — so we can do geometry in $\mathbb{R}^n$. (For this we need Cauchy–Schwarz!)

Step 1 — geometric definition (Def. 2.1.1)

For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ (or $\mathbb{R}^2$), with $\varphi \in [0, \pi]$ the angle between them:

$$\mathbf{v} \cdot \mathbf{w} := |\mathbf{v}| |\mathbf{w}| \cos \varphi, \quad \text{and } \mathbf{v} \cdot \mathbf{w} := 0 \text{ if } \mathbf{v} = \mathbf{0} \text{ or } \mathbf{w} = \mathbf{0}.$$

Meetkundig: drop the perpendicular from the head of $\mathbf{w}$ onto the line of $\mathbf{v}$; call the foot vector $\mathbf{x}$ (the *loodrechte projectie* of $\mathbf{w}$ on $\mathbf{v}$). Then $\cos \varphi = |\mathbf{x}|/|\mathbf{w}|$, so $|\mathbf{x}| = |\mathbf{w}| \cos \varphi$ and

$$\mathbf{v} \cdot \mathbf{w} = |\mathbf{v}| |\mathbf{x}|.$$

So the inner product is “length of $\mathbf{v}$ times the length of the shadow of $\mathbf{w}$ on $\mathbf{v}$ ”.

Three immediate consequences (“we zien”):

1. $\mathbf{v} \cdot \mathbf{w} = 0 \iff \mathbf{v} \perp \mathbf{w}$ or $\mathbf{v} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$.
2. $\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2 \geq 0$, and $\mathbf{v} \cdot \mathbf{v} = 0 \iff \mathbf{v} = \mathbf{0}$.
3. If $\mathbf{v}, \mathbf{w} \neq \mathbf{0}$ then $\varphi = \cos^{-1} \left(\frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}| |\mathbf{w}|} \right)$.

Point 3 is only useful if we can compute $\mathbf{v} \cdot \mathbf{w}$ *without* already knowing φ . That’s step 2.

Step 2 — in coordinates (Stelling 2.2.2)

If $\mathbf{x} = (x_1, x_2, x_3)^t$ and $\mathbf{y} = (y_1, y_2, y_3)^t$ then

$$\boxed{\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + x_3 y_3}$$

Bewijsschets (as in the lecture). Draw the triangle with sides $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{y} - \mathbf{x}$. The **cosinusregel** (Stelling 2.2.1) gives

$$|\mathbf{y} - \mathbf{x}|^2 = |\mathbf{x}|^2 + |\mathbf{y}|^2 - 2 |\mathbf{x}| |\mathbf{y}| \cos \varphi = |\mathbf{x}|^2 + |\mathbf{y}|^2 - 2 \mathbf{x} \cdot \mathbf{y}.$$

Hence

$$\mathbf{x} \cdot \mathbf{y} = \frac{1}{2} (|\mathbf{x}|^2 + |\mathbf{y}|^2 - |\mathbf{y} - \mathbf{x}|^2),$$

and the right-hand side you simply write out in coordinates using the length formula; all the squares cancel and $x_1 y_1 + x_2 y_2 + x_3 y_3$ remains (full computation: dictaat p. 20).

Step 3 — definition on $\mathbb{R}^n$ (Def. 2.5.2)

For $\mathbf{x} = (x_1, \dots, x_n)^t$, $\mathbf{y} = (y_1, \dots, y_n)^t$ we **define**

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n, \quad |\mathbf{x}| = \sqrt{\mathbf{x} \cdot \mathbf{x}}.$$

The algebraic properties (symmetry, scalars pull out, distributivity, $\mathbf{x} \cdot \mathbf{x} \geq 0$) are Stelling 2.2.4 (lecture slides 8–11).

Opmerking over Cauchy–Schwarz (Stelling 2.5.3):

$$|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| |\mathbf{y}|.$$

- For $n = 2, 3$ this is immediate: $|\mathbf{x} \cdot \mathbf{y}| = |\mathbf{x}| |\mathbf{y}| \cos \varphi \leq |\mathbf{x}| |\mathbf{y}|$.
- For $n > 3$ there is no picture; the proof is in the dictaat (p. 25) and is “niet intuïtief”.
- **Gevolg:** because Cauchy–Schwarz guarantees $\frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}| |\mathbf{y}|} \in [-1, 1]$, we can *define* the angle between $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ (both $\neq \mathbf{0}$) as $\varphi = \cos^{-1} \left(\frac{\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}| |\mathbf{y}|} \right)$. That is what “geometry in $\mathbb{R}^n$ ” means.

7. Equation of a plane in $\mathbb{R}^3$ via the inner product (§2.3)

Let V be a plane with steunvector $\mathbf{p}$.

Def. (2.3.1): $\mathbf{n} \neq \mathbf{0}$ is a **normaalvector** of V if $\mathbf{n}$ is perpendicular to V .

What does “perpendicular to V ” mean? For every $\mathbf{x}$ on V the vector $\mathbf{x} - \mathbf{p}$ lies *in* V (picture: all the red arrows from $\mathbf{p}$ within the plane), so

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0 \quad \text{for all } \mathbf{x} \in V.$$

Expand: $\mathbf{n} \cdot \mathbf{x} - \mathbf{n} \cdot \mathbf{p} = 0$, i.e.

$$\boxed{\mathbf{n} \cdot \mathbf{x} = \mathbf{n} \cdot \mathbf{p}} \quad n_1x_1 + n_2x_2 + n_3x_3 = (\text{a number}).$$

This is the **vergelijking van het vlak** (Def. 1.4.5: $ax_1 + bx_2 + cx_3 = d$).

VB (lecture): $\mathbf{p} = (0, 1, 1)^t$, $\mathbf{n} = (1, 1, 1)^t$. Then $\mathbf{x} \in V \iff (1, 1, 1)^t \cdot \mathbf{x} = (1, 1, 1)^t \cdot (0, 1, 1)^t = 2$, so

$$x_1 + x_2 + x_3 = 2.$$

Two descriptions of the same plane

Parametrisatie	Vergelijking
$V = \{\mathbf{p} + \lambda\mathbf{a} + \mu\mathbf{b} \mid \lambda, \mu \in \mathbb{R}\}$	$(\mathbf{x} - \mathbf{p}) \cdot \mathbf{n} = 0$

Omgekeerd: the coefficients *are* a normal vector

Claim. If $ax_1 + bx_2 + cx_3 = d$ is an equation of a plane V , then $\mathbf{n} = (a, b, c)^t$ is a normaalvector of V .

Bewijs (from the notes). Let $\mathbf{p} = (p_1, p_2, p_3)^t$ be a steunvector (so $ap_1 + bp_2 + cp_3 = d$). We want: for every $\mathbf{x} \in V$, $(\mathbf{p} - \mathbf{x}) \perp \mathbf{n}$. Compute

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} p_1 - x_1 \\ p_2 - x_2 \\ p_3 - x_3 \end{pmatrix} = (ap_1 + bp_2 + cp_3) - (ax_1 + bx_2 + cx_3) = d - d = 0,$$

so $(\mathbf{p} - \mathbf{x}) \perp \mathbf{n}$. $\square$

From equation to parametrisation (slide vraag 5)

Plane V : $3x_1 + 4x_2 + x_3 = 2$. Solve for one variable: $x_3 = 2 - 3x_1 - 4x_2$. Let $x_1 = \lambda$, $x_2 = \mu$ be free:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \lambda \\ \mu \\ 2 - 3\lambda - 4\mu \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix},$$

a parametrisation of V . (Check: $(1, 0, -3) \cdot (3, 4, 1) = 0$ and $(0, 1, -4) \cdot (3, 4, 1) = 0$, so both directions are indeed $\perp \mathbf{n}$.)

From parametrisation to equation you go the other way: eliminate λ, μ , or find $\mathbf{n}$ perpendicular to both direction vectors and use $\mathbf{n} \cdot \mathbf{x} = \mathbf{n} \cdot \mathbf{p}$ (Voorbeeld 2.3.2 in the dictaat).

Zelftest (the lecture's slide questions)

1. **Vraag 1/2.** Give a parametrisation of the line through $P = (1, 2, -1)^t$ and $Q = (2, 0, 1)^t$.
Answer: $\{\mathbf{p} + \lambda(\mathbf{q} - \mathbf{p})\} = \{(1, 2, -1)^t + \lambda(1, -2, 2)^t \mid \lambda \in \mathbb{R}\}$ (any steunvector on the line and any multiple of the direction is fine).
2. **Vraag 3.** Parametrise the plane through P, Q, R . *Answer:* $\{\mathbf{p} + \lambda(\mathbf{q} - \mathbf{p}) + \mu(\mathbf{r} - \mathbf{p}) \mid \lambda, \mu \in \mathbb{R}\}$, valid when P, Q, R are not collinear.
3. **Vraag 4.** Angle between $(1, 1, 1)^t$ and $(1, 0, 0)^t$. *Answer:* $\mathbf{x} \cdot \mathbf{y} = 1$, $|\mathbf{x}| = \sqrt{3}$, $|\mathbf{y}| = 1$, so $\varphi = \cos^{-1}(1/\sqrt{3})$. (This was the “blauw” answer on the slide.)
4. **Vraag 5.** Parametrise $3x_1 + 4x_2 + x_3 = 2$. *Answer:* see §7 above.
5. Why does $\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2$? *Answer:* $\varphi = 0$, $\cos 0 = 1$.
6. A plane has equation $x_1 - 2x_2 + 5x_3 = 7$. Write down a normal vector and one point on it.
Answer: $\mathbf{n} = (1, -2, 5)^t$; e.g. $\mathbf{p} = (7, 0, 0)^t$.